SQL • POWER BI • PYTHON • MACHINE LEARNING

Online Shoppers Purchasing Intention Analysis & Prediction

An end-to-end data analytics and machine learning project predicting purchase intent from 12,330 e-commerce browsing sessions

SQL Analysis

EDA & Dashboard

ML Experiments

Final Model

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ROADMAP

What We'll Cover Today

Business Problem & Dataset

Objectives, scope, and the 18-feature session-level dataset

Power BI Dashboard

Interactive KPI dashboard for stakeholders

Data Cleaning & Preprocessing

Duplicate removal, missing values, outlier treatment

Machine Learning Experiments

4 experiments: Logistic Regression, transformations, class weighting, tree models

Exploratory Data Analysis

Univariate, bivariate, and correlation insights

Final Model & Recommendations

Model comparison, ranking, and business recommendations

SQL Business Analysis

18 business questions answered directly in SQL

BUSINESS UNDERSTANDING

Problem Statement & Objectives

The Problem

E-commerce businesses often struggle to understand why some users purchase products while others leave the website without converting. Most visitors browse and exit without buying, and businesses need to know which behavioral signals separate buyers from non-buyers.

The goal is to build a machine learning model that predicts whether a visitor is likely to make a purchase, based on browsing behavior, session characteristics, and website interaction data.

Predict Purchase Intent

Classify sessions as revenue-generating or not, using browsing signals

Target High-Intent Visitors

Help marketing teams personalize offers for likely buyers

Improve Conversion Rate

Turn behavioral insight into concrete conversion-boosting actions

Reduce Wasted Spend

Identify low-quality traffic and disengaged segments to cut costs

THE DATA

Dataset Overview

12,330	18	10	15.63%
Total Sessions	Original Features	Months Covered	Overall Conversion Rate

Each row represents one customer browsing session on an e-commerce site. Features fall into four business-meaningful groups:

User Behavior

Administrative, Informational & ProductRelated page counts and durations

Interaction Metrics

BounceRates, ExitRates, PageValues, SpecialDay proximity

Visitor Information

VisitorType (New/Returning), Region, Browser, OperatingSystem, TrafficType

Session Context & Target

Month, Weekend flag, and the target — Revenue (purchase made or not)

Data Cleaning & Preprocessing

1. **Shape Check:** 12,330 rows × 18 features confirmed
2. **Duplicate Removal:** 125 exact duplicates removed — identical floats impossible for real users
3. **Missing Values:** Zero missing values across all 18 columns
4. **Data Type Fix:** Revenue & Weekend converted from bool → int
5. **Month Fix:** 'June' → 'Jun' for consistency
6. **Validation:** BounceRate & ExitRate within [0,1] — no invalid values
7. **Feature Engineering:** TotalPages & TotalDuration created from sub-features

```
# Remove exact duplicates
df.drop_duplicates(inplace=True)
print("Shape:", df.shape)  # (12205, 18)

# Fix dtypes
df["Revenue"] = df["Revenue"].astype(int)
df["Weekend"] = df["Weekend"].astype(int)

# Fix month inconsistency
df["Month"] = df["Month"].replace("June","Jun")

# Feature engineering
df['TotalPages'] = (df['Administrative']
  + df['Informational']
  + df['ProductRelated'])

df['TotalDuration'] = (
  df['Administrative_Duration']
  + df['Informational_Duration']
  + df['ProductRelated_Duration’])

# Missing value scan
df.isnull().sum()
```

125 Duplicates Removed	**12,205** Clean Sessions	**0** Missing Values	**2** New Features Added

EXPLORATORY DATA ANALYSIS

Univariate Analysis — Key Findings

Categorical variables (Region, Browser, TrafficType, Weekend) were analyzed by frequency distribution; numerical variables were analyzed via histograms and boxplots.

Product Pages Dominate Engagement

Users spend 1,206 seconds on average on product pages — dwarfing all other page types. That's where the entire conversion opportunity lives.

Returning Visitors Are 77% of Traffic

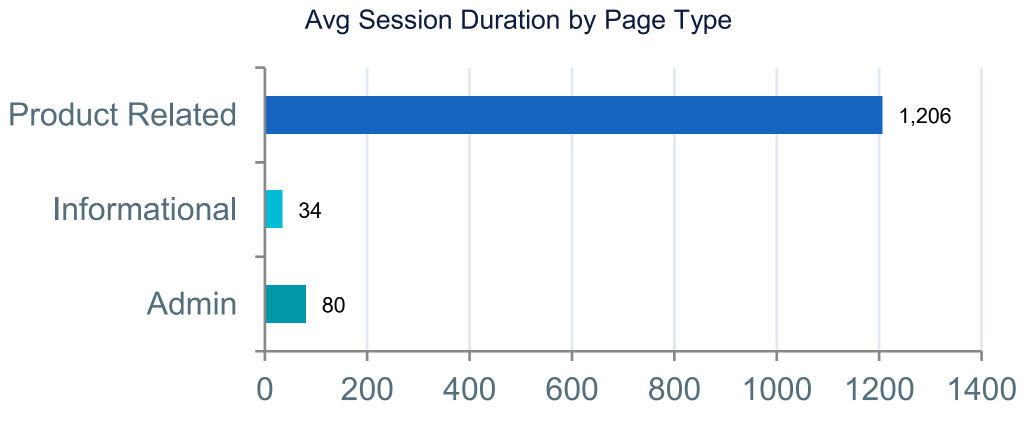
The site has a massive loyal base that barely converts. Even a small improvement here means a big revenue impact.

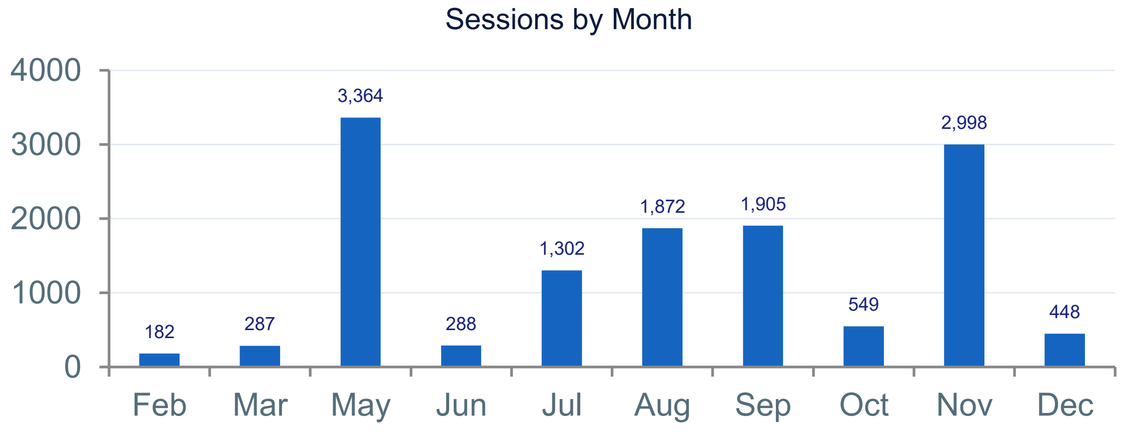
PageValues Are Near-Zero for Most Sessions

Almost all sessions contribute no revenue signal at all — purchases are driven by a small fraction of high-PageValue sessions.

May & November Are the Only High-Traffic Months

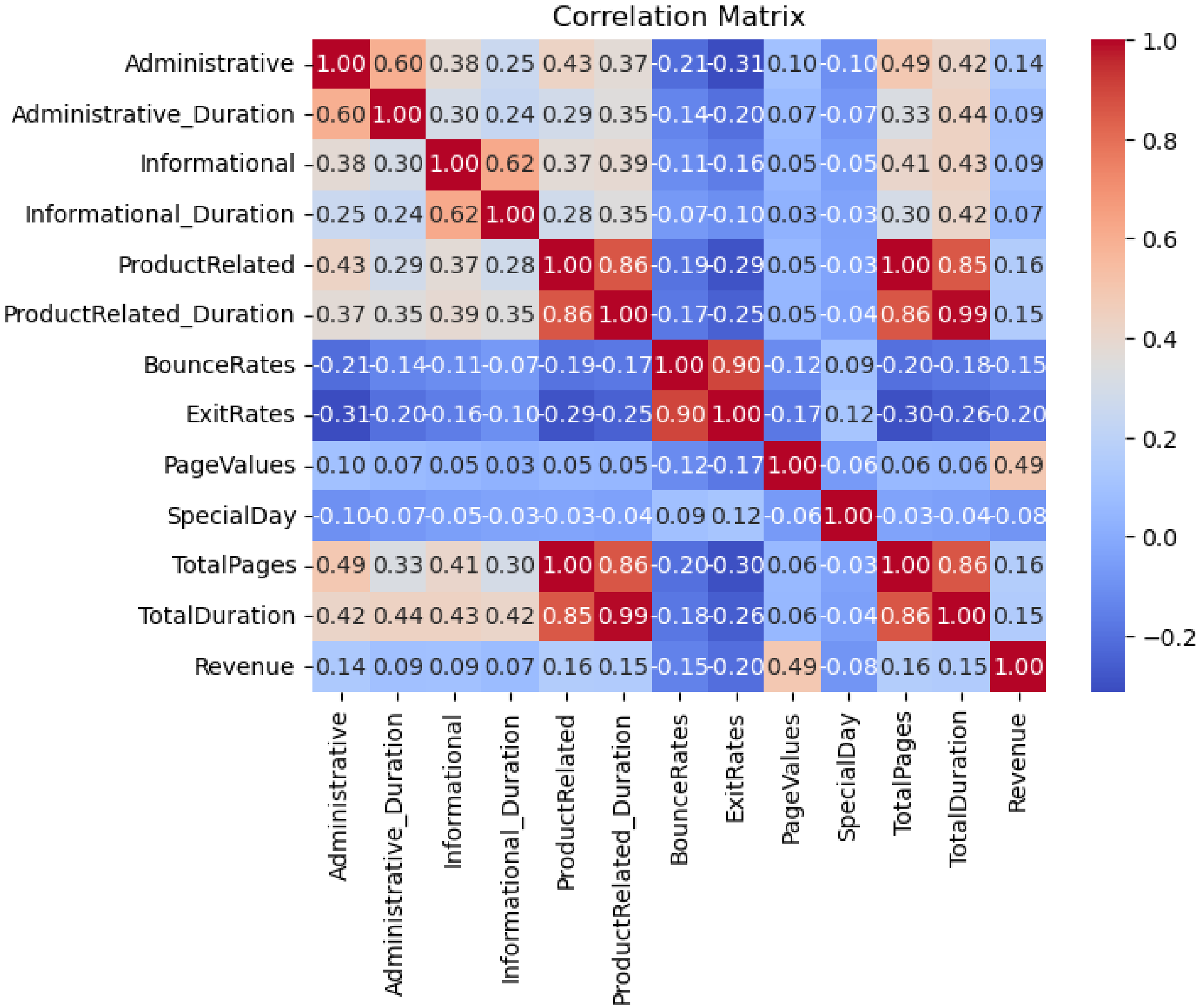
These two months alone dominate overall session volume across the year.





Correlation Analysis

Numerical Feature Correlation Heatmap



BounceRates ↔ ExitRates **0.90**

Very strong positive correlation — a visitor who bounces on one page tends to exit quickly elsewhere too.

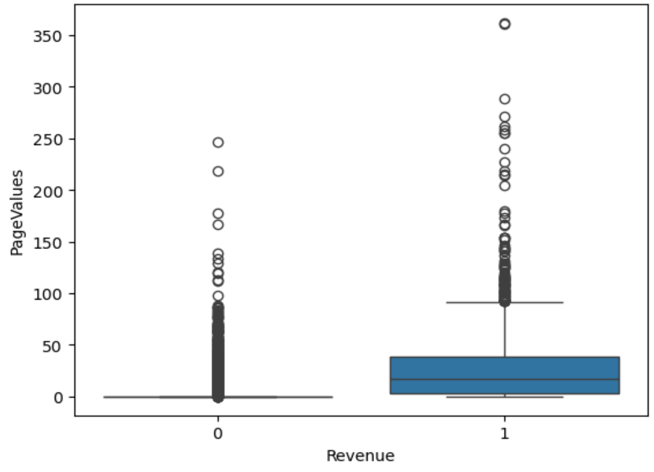
ProductRelated ↔ ProductRelated_Duration **0.86**

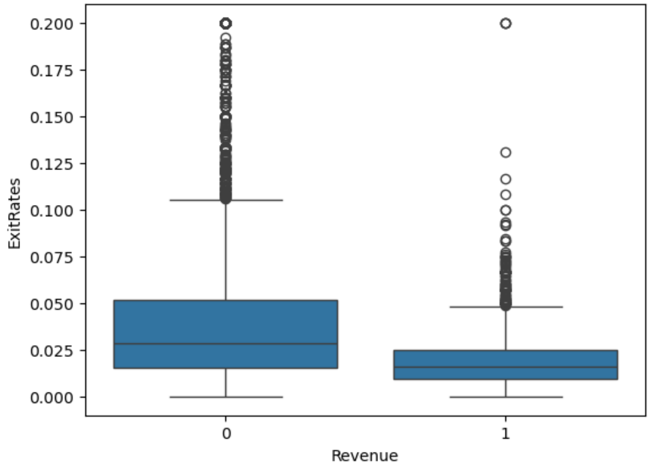
Strong positive correlation — more product pages viewed naturally means more time spent browsing them.

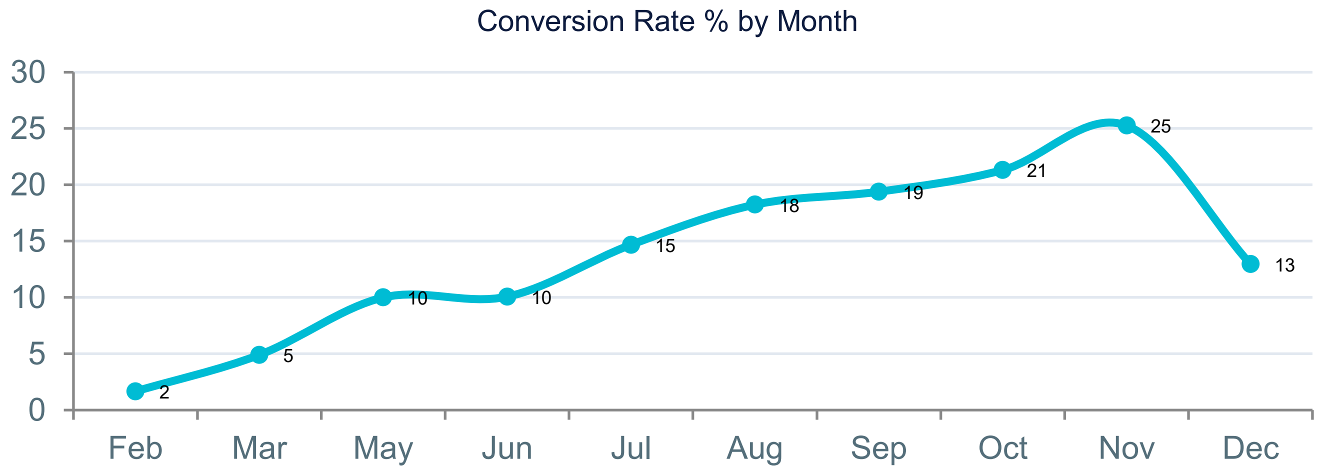
PageValues ↔ Revenue **High**

Sessions with higher page values are far more likely to convert — the strongest revenue signal in the data.

Bivariate Analysis & Key Findings







KEY FINDING SUMMARY

- PageValues of buyers are 13.6x higher than non-buyers.
- Buyers have 4.5x lower bounce rates than non-buyers.
- Only 15.63% of visitors actually buy — 84.37% browse and leave
- Buyers spend 73% more time on product pages (1,876s vs 1,083s for non-buyers)
- New visitors convert at 24.9% vs returning visitors at only 14.1%
- November converts best (25.49%), February worst (1.66%) — a 15x seasonal swing

The Story the Data Tells

15.63%

Overall conversion rate — 84.37% of visitors browse and leave without buying

73% more

time spent on product pages by buyers (1,876s) vs non-buyers (1,083s)

15x

difference between best month, November (25.49%), and worst, February (1.66%)

2x

New visitors convert at 24.9% vs returning visitors at just 14.1%

4.5x

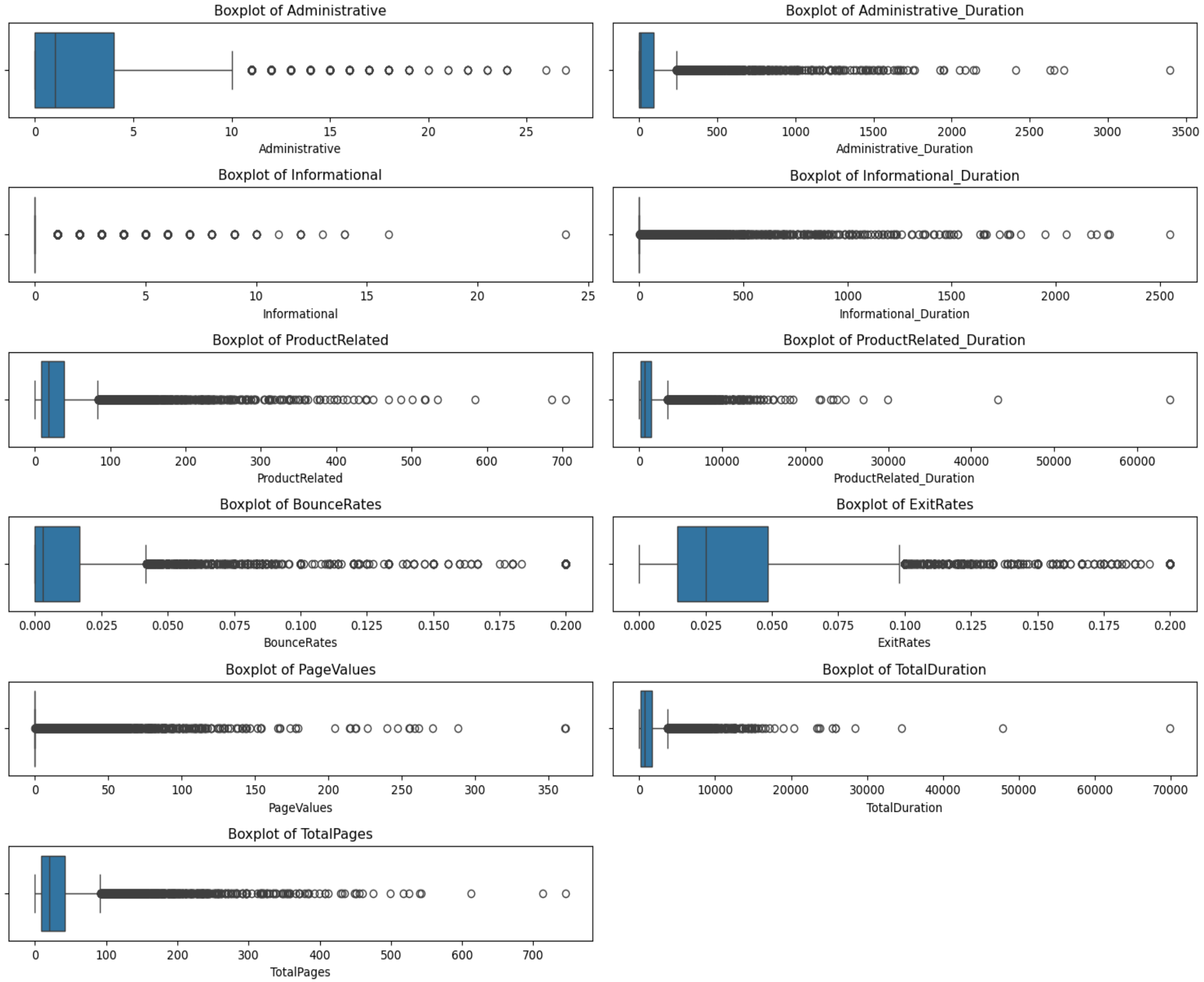
lower bounce rate for buyers (0.005) vs non-buyers (0.023)

13.6x

higher PageValues for buyers (27.26) vs non-buyers (2.0)

PREPROCESSING

Outlier Detection (IQR Method)



IQR METHOD

```
Q1 = df[col].quantile(0.25)
Q3 = df[col].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
outliers =
df[(df[col]<lower)|(df[col]>upper)]
```

WHY OUTLIERS EXIST

- Administrative / Informational pages: a small group of power-users visit repeatedly (order history, FAQs, policies) creating high variance
- ProductRelated & Duration — comparison shopping or idle browser tabs left open
- BounceRates — irrelevant traffic or accidental clicks causing instant exits
- ExitRates — checkout abandonment or comparison browsing before leaving
- PageValues — a small group of high-value customers on high-converting pages
- **Decision**: outliers were retained (not removed) since they represent real high-value buyer behaviour critical to the target variable

Answering Business Questions in SQL

18 business questions were answered directly in SQL against the cleaned online_shoppers table — each paired with an insight and an actionable recommendation.

SQL — CONVERSION RATE BY MONTH

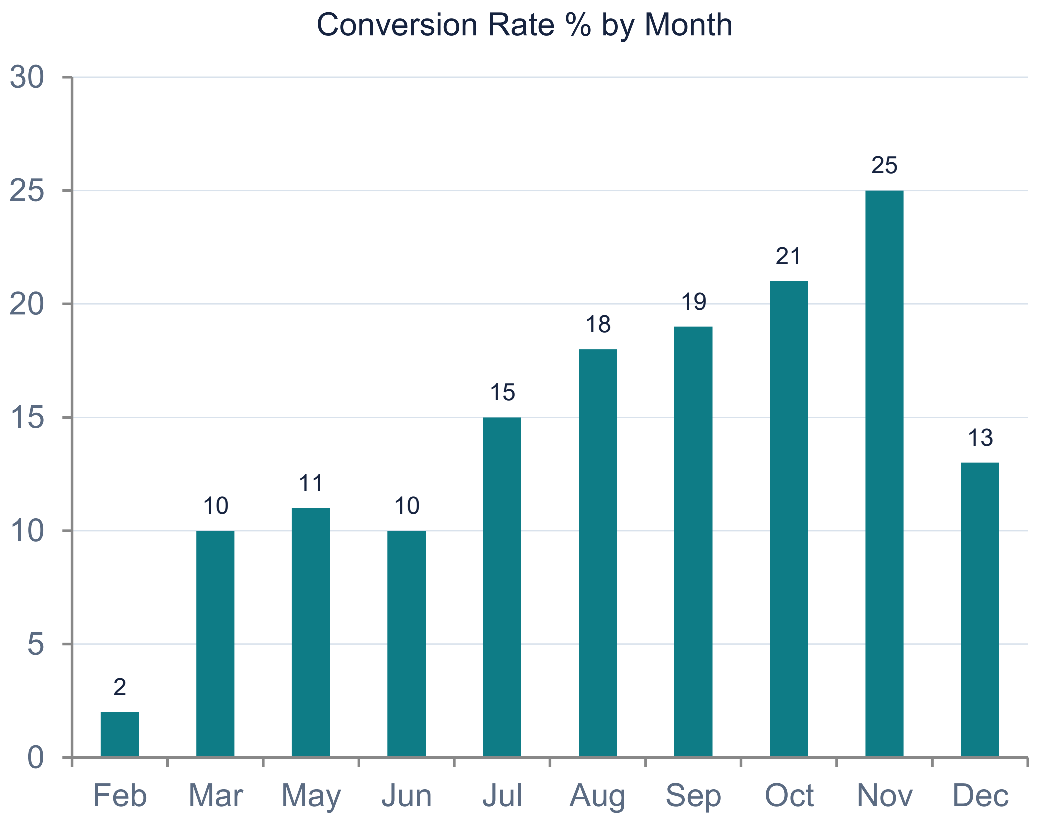
```
SELECT Month,
       ROUND(AVG(Revenue) * 100, 2)
         AS Conversion_Rate
FROM project.online_shoppers
GROUP BY Month
ORDER BY Conversion_Rate DESC;
```

INSIGHT

November and October recorded the highest conversion rates, showing strong seasonal purchase intent during festive/sale periods.

RECOMMENDATION

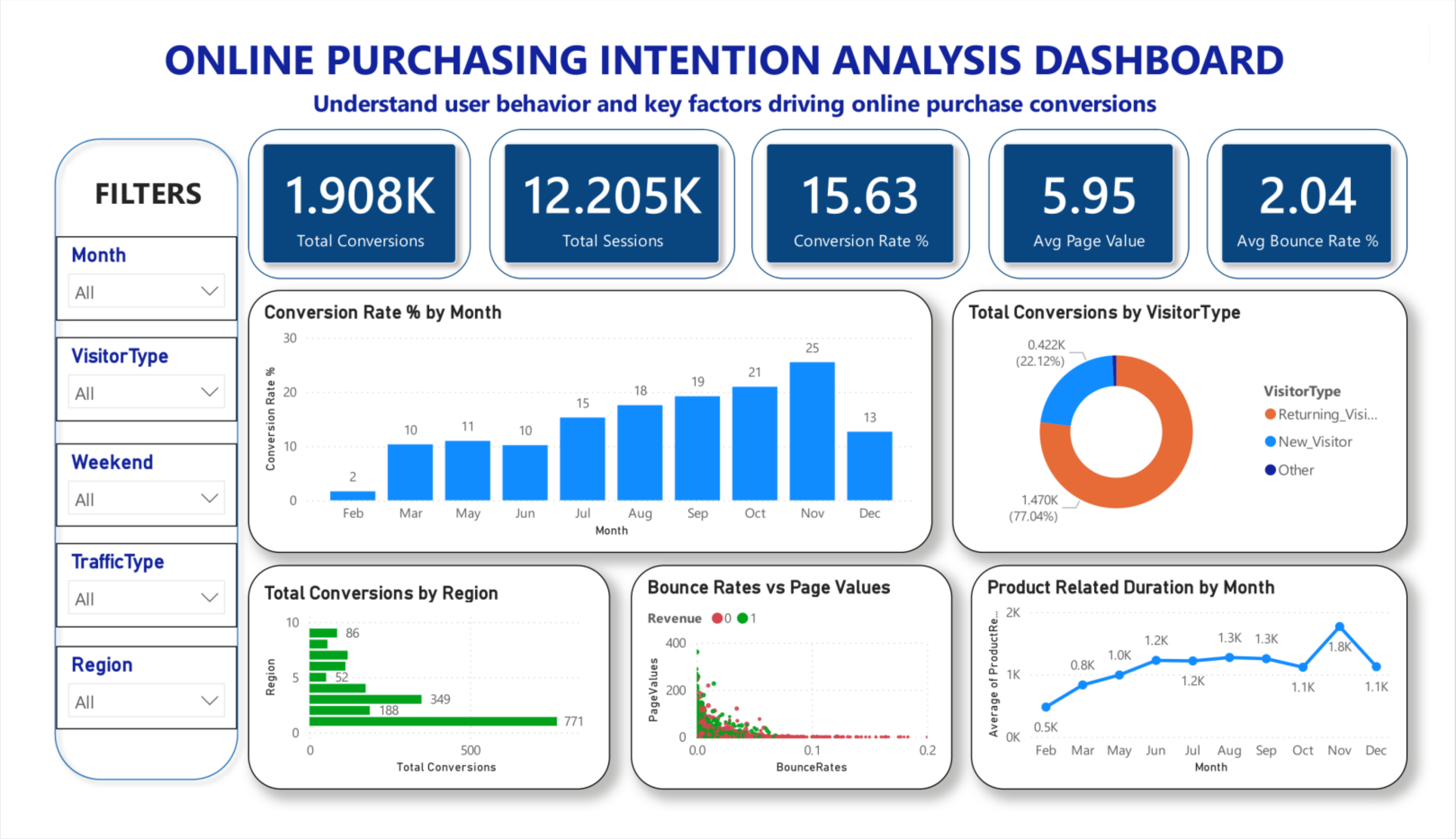
Increase marketing campaigns and promotional offers during high-conversion months to maximize revenue.



More SQL-Driven Insights

Business Question	Key Insight	Recommendation
Which visitor type generates highest page value?	New Visitors generate higher PageValues per session than Returning Visitors.	Focus on converting new visitors with personalized onboarding.
Which traffic sources convert best / worst?	Traffic Types 16, 7, 8 perform best; several sources have near-zero conversions.	Invest in high-performing channels; optimize or cut weak ones.
Do bounce rates affect conversion?	Converted users have much lower bounce rates than non-converters.	Optimize landing pages and ad targeting to reduce bounce behavior.
Do weekend visitors convert better?	Weekend visitors show slightly higher conversion rates than weekdays.	Increase weekend promotions and targeted campaigns.
Which regions contribute highest revenue?	Regions 9, 2, 1, and 5 show the highest conversion rates; Region 8 lowest.	Focus regional marketing on top regions; improve weak markets.
Does SpecialDay proximity drive purchases?	Conversion is actually highest when SpecialDay = 0 (regular periods).	Don't rely only on holiday promos — sustain year-round engagement.

Interactive KPI Dashboard



Filters: Month · VisitorType · Weekend · TrafficType · Region — enabling stakeholders to slice conversion behavior interactively.

KPI Cards (5)

Total Conversions (1.908K) · Total Sessions (12.205K) · Conversion Rate % (15.63) · Avg Page Value (5.95) · Avg Bounce Rate % (2.04)

Conversion Rate by Month (Bar)

Column chart showing monthly trend — Nov peaks at 25%. Used to identify seasonal revenue opportunities.

Total Conversions by Visitor Type (Donut)

77.04% Returning Visitors · 22.12% New Visitors. Reveals the paradox: Returning dominate traffic but New Visitors convert better.

Total Conversions by Region (Bar)

Horizontal bar: Region 1 leads with 771 conversions, Region 2 with 349. Supports regional campaign planning.

Bounce Rate vs Page Values (Scatter)

Revenue-coded scatter: Green dots (purchases) cluster at low bounce & high page value. Visually proves the engagement-conversion correlation.

Product Duration by Month (Line)

Line chart of avg product browsing duration — Nov highest at 1.8K sec. Guides campaign timing for content-heavy months.

MACHINE LEARNING

Four Experiments, One Objective

From a simple linear baseline to a fully tuned ensemble model — systematically improving Recall & F1-Score on an imbalanced target.

EXP 1

Baseline Logistic Regression

Establish a benchmark with raw scaled features

EXP 2

Log1p & Yeo-Johnson + LR

Reduce skewness in duration/value features

EXP 3

Transform + Class-Weighted LR

Address class imbalance (only 15.6% positive)

EXP 4

Tuned Decision Tree & Random Forest

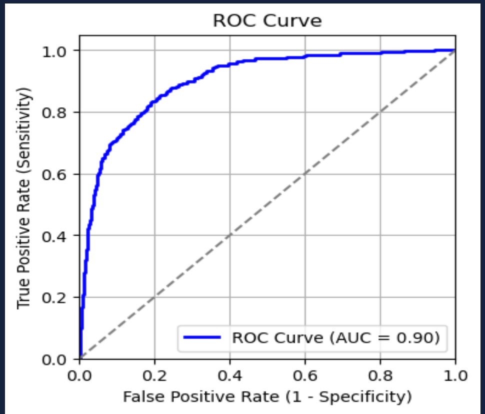
Capture non-linear patterns; tune with search CV

EXPERIMENT 1

Baseline Logistic Regression

plain sklearn LogisticRegression, no class balancing, no transformation -- purely to see how a linear model performs “out of the box” on an imbalanced 84:16 target

- Encoding: pd.get_dummies() on Month & VisitorType (drop_first=True)
- Split: 80/20 train-test, stratified on Revenue to preserve class ratio
- Scaling: RobustScaler (robust to the outliers we chose to keep)
- Model: plain sklearn LogisticRegression, no class balancing, no transformation



Test ROC-AUC = 0.902

Confusion Matrix (Test Set)

	Pred: No	Pred: Yes
Actual: No	2,012	47
Actual: Yes	223	159

89%	0.77	0.42	0.54	0.902
Test Accuracy	Precision	Recall	F1-Score	ROC-AUC

Takeaway

The baseline model serves as the benchmark. It achieved the highest Precision of the five experiments (0.77) but a very low Recall (0.42) — meaning it missed a large share of actual buyers. This motivated exploring feature transformations and class weighting in later experiments.

Why it under-detects buyers

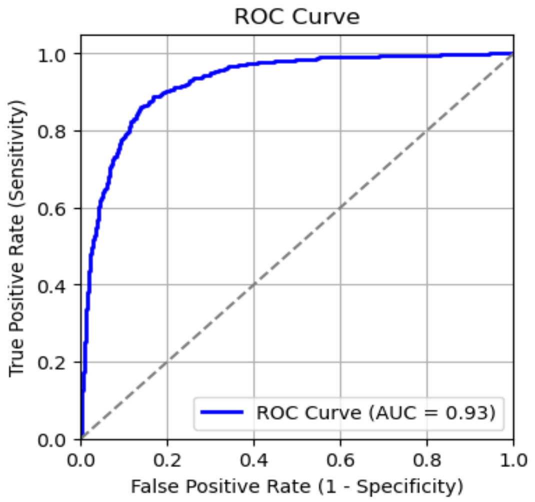
With only 15.6% of sessions converting, a plain Logistic Regression naturally leans toward predicting the majority class (“no purchase”) — which is exactly the imbalance problem Experiments 2–4 address.

EXPERIMENT 2

Fixing Skewness — Log1p vs Yeo-Johnson

Numeric features (durations, PageValues) were heavily right-skewed. Two transformations were tested to normalise them before feeding into Logistic Regression.

Metric	Log1p + LR	Yeo-Johnson + LR
Test Accuracy	**90.29%**	89.79%
Precision	**0.72**	0.70
Recall	**0.61**	0.60
F1-Score	**0.66**	0.65
Test ROC-AUC	**0.927**	0.920



TEST ROC — Log1p + LR

WINNER

Log1p

Wins on Accuracy, Precision, Recall, F1 and ROC-AUC — simpler transform, better outcome.

Conclusion: Although Yeo-Johnson reduced skewness more, Log1p gave better predictive performance on every metric. Log1p was carried forward into Experiment 3.

EXPERIMENT 3

Handling Class Imbalance — class_weight='balanced'

Only 15.63% of sessions convert. Instead of resampling (SMOTE), class_weight='balanced' was used inside Logistic Regression to penalise misclassifying the minority (buyer) class more heavily.

```
lr = LogisticRegression(
    class_weight='balanced',
    random_state=42,
    max_iter=1000)
lr.fit(X_train_scaled, y_train)
```

Metric	Log1p+CW	Yeo-J+CW
Accuracy	**87%**	86%
Precision	**0.56**	0.54
Recall	0.81	**0.83**
F1-Score	**0.66**	0.65
ROC-AUC	**0.929**	0.923

RECALL JUMPS FROM 0.61 → 0.81

Class weighting makes the model far more sensitive to buyers — it now catches 80% of real purchasers (vs 42% in the baseline). The trade-off: more false positives, so Precision drops to 0.56 and Accuracy to 87%.

When to use this model: when the cost of missing a real buyer is higher than the cost of contacting a few extra non-buyers (e.g. cart-abandonment email campaigns).

Conclusion: Log1p + Class Weight selected over Yeo-Johnson + Class Weight — better Accuracy, Precision, F1 & ROC-AUC, with Recall only marginally lower (0.81 vs 0.83).

EXPERIMENT 4A

Decision Tree — Default vs GridSearchCV Tuned

Log1p-transformed features carried forward. A default Decision Tree overfit badly (100% train accuracy) — GridSearchCV was used to tune it.

```
param_grid = {
  'max_depth': [3,5,7,10,None],
  'min_samples_split': [2,5,10,20],
  'min_samples_leaf': [1,2,5,10],
  'criterion': ['gini','entropy'],
  'max_features': [None,'sqrt','log2']}

grid_dt = GridSearchCV(dt, param_grid, cv=5,
    scoring='f1', n_jobs=-1)
# 480 candidates x 5 folds = 2400 fits
```

Best Params: criterion='gini', max_depth=7, min_samples_leaf=5, min_samples_split=20 | Best CV F1 = 0.642

Metric	Default DT	Tuned DT
Train Accuracy	1.00 (overfit)	0.87
Test Accuracy	86.03%	**85.25%**
Precision	0.55	0.52
Recall	0.55	0.81
F1-Score	0.55	0.63

Tuning removed the massive train/test gap (overfitting) and boosted Recall to 0.81, though it stays behind Logistic Regression + Class Weight overall.

EXPERIMENT 4B

Random Forest — Default vs RandomizedSearchCV Tuned

```
param_dist = {
  'n_estimators': [100,200,300,500],
  'max_depth': [5,10,15,20,None],
  'min_samples_split': [2,5,10,20],
  'min_samples_leaf': [1,2,5,10],
  'max_features': ['sqrt','log2'],
  'bootstrap': [True, False]}

random_rf = RandomizedSearchCV(rf, param_dist,
    n_iter=30, cv=5, scoring='f1', n_jobs=-1)
# 30 candidates x 5 folds = 150 fits
```

Best Params: n_estimators=300, max_depth=15, min_samples_split=10, min_samples_leaf=2, max_features='log2' | Best CV F1 = 0.686

Metric	Default RF	Tuned RF
Train Accuracy	1.00 (overfit)	**0.947**
Test Accuracy	90.20%	**89.34%**
Precision	0.76	0.64
Recall	0.55	**0.75**
F1-Score	0.64	**0.69**
ROC-AUC (test)	0.924	**0.928**

Best model of all 4 experiments — highest F1 (0.69) and strong ROC-AUC (0.929), with balanced Precision & Recall.

FINAL COMPARISON

All Five Experiments, Side by Side

Rank	Experiment	Accuracy	Precision	Recall	F1	ROC-AUC
1	**Random Forest (tuned) — FINAL MODEL**	89%	0.64	0.75	**0.69**	**0.928**
2	Logistic Regression + Log1p	90.29%	0.72	0.61	0.66	0.926
3	Logistic Regression + Log1p + Class Weight	87%	0.56	0.81	0.66	0.929
4	Decision Tree (tuned)	85%	0.52	0.81	0.63	0.919
5	Logistic Regression (baseline)	89%	0.77	0.42	0.54	0.902

Final Recommendation: Tuned Random Forest

Although Log1p Logistic Regression achieved the highest raw accuracy (90.29%), accuracy alone is not sufficient on this imbalanced dataset. The tuned Random Forest offers the best overall balance — highest F1-score (0.69) and excellent ROC-AUC (0.929) — making it the most suitable model for production use in identifying real buyers while minimizing missed opportunities.

BUSINESS IMPACT

Recommendations & Conclusion

Build a Seasonal Calendar

Plan campaigns, stock, and offers from September onward instead of waiting for November; use Feb–Mar for brand awareness and list-building.

Cut Low-Quality Traffic

Traffic types with near-zero conversion and high bounce rates (15, 17, 18) should be re-targeted or de-prioritized in ad spend.

Re-engage Returning Visitors

Returning visitors are 77% of traffic but convert less than new visitors — loyalty offers and remarketing can close this gap.

Invest in Product Page

Add detailed descriptions, reviews, comparison tools, and smart related-product sections. This is where the purchase decision is made.

The data tells a clear story: engaged users buy, disengaged users don't. A production-ready Random Forest model, paired with these seasonal and segment-level actions, gives the business a concrete path to lift the 15.63% conversion rate.

github.com/komalkakkar1996-source